

NAGORNY, Russian FederationWMO: **304930**Lat: **55.948N**Lon: **124.915E**Elev: **861**StdP: **91.40**Time Zone: **9.00 (E09)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-41.4	-39.1	-43.5	0.1	-41.0	-41.4	0.1	-38.7	9.4	-28.4	8.2	-26.9	0.8	270	0.715

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.9	26.4	16.3	24.5	15.7	22.6	14.8	18.1	22.7	17.1	21.5	16.2	20.6	1.4	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.8	13.3	19.1	15.7	12.4	18.5	14.6	11.5	17.7	54.8	22.6	51.7	21.3	48.8	20.5	24.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.7	5.5	4.6	-45.1	29.2	2.1	1.8	-46.5	30.5	-47.7	31.5	-48.9	32.5	-50.4	33.8		
(5)			-44.9	19.0	1.9	2.5	-46.2	20.8	-47.4	22.3	-48.4	23.7	-49.8	25.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-6.7	-28.3	-24.5	-16.3	-5.5	4.6	12.8	15.2	12.6	4.4	-7.4	-21.1	-28.0		
	DBStd	16.67	5.64	5.32	5.93	5.24	4.05	3.64	2.85	3.39	4.34	5.83	7.24	5.58		
	HDD10.0	6462	1186	967	817	464	173	15	2	14	173	541	934	1178		
	HDD18.3	9146	1444	1201	1075	714	427	169	102	178	418	799	1184	1436		
	CDD10.0	367	0	0	0	0	4	98	164	95	6	0	0	0		
	CDD18.3	9	0	0	0	0	0	2	6	1	0	0	0	0		
	CDH23.3	292	0	0	0	0	2	98	143	49	1	0	0	0		
	CDH26.7	43	0	0	0	0	0	14	25	4	0	0	0	0		
	WSAvg	1.5	1.8	1.6	1.8	1.8	1.7	1.2	1.1	1.0	1.4	1.6	1.4	1.5		
(15) Precipitation	PrecAvg	568	7	6	11	31	40	101	135	116	73	25	14	9		
	PrecMax	721	21	18	44	116	99	212	231	214	145	69	36	24		
	PrecMin	370	0	0	1	10	3	14	25	32	6	4	2	2		
	PrecStd	94	5	4	10	20	25	54	50	46	31	14	9	6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-10.9	-7.6	1.8	12.7	22.9	28.5	29.5	26.8	20.5	8.9	-0.2	-11.3		
		MCWB	-12.1	-9.2	-1.6	5.8	11.8	16.4	17.8	17.3	12.5	3.9	-2.3	-12.2		
	2%	DB	-15.0	-10.8	-1.3	8.3	19.0	26.1	26.7	24.7	17.4	5.3	-4.2	-14.3		
		MCWB	-15.8	-12.0	-3.8	2.8	9.8	15.4	16.8	16.3	10.8	1.7	-5.8	-15.0		
	5%	DB	-17.9	-13.6	-3.8	5.3	15.9	23.9	24.7	22.5	14.7	2.7	-7.0	-16.8		
		MCWB	-18.5	-14.4	-5.9	0.9	8.3	14.7	16.4	15.4	9.2	0.0	-8.2	-17.4		
	10%	DB	-20.0	-16.1	-6.1	3.2	13.2	21.8	22.7	20.3	12.2	0.7	-10.0	-19.5		
		MCWB	-20.5	-16.8	-7.9	-0.6	7.0	13.6	15.6	14.5	8.5	-1.2	-10.8	-19.9		
	0.4%	WB	-12.0	-9.1	-1.4	6.1	12.6	18.5	19.9	19.3	13.9	4.2	-2.1	-12.2		
		MCDB	-10.8	-7.7	1.2	12.0	20.7	23.7	25.6	23.3	18.7	8.1	0.0	-11.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	-15.8	-11.9	-3.6	3.1	10.4	17.1	18.5	17.4	11.9	2.0	-5.7	-15.0		
		MCDB	-15.1	-10.8	-1.4	8.2	17.5	23.2	23.4	21.5	15.2	4.8	-4.1	-14.3		
	5%	WB	-18.4	-14.4	-5.9	1.0	8.9	15.9	17.5	16.5	10.5	0.3	-8.1	-17.4		
		MCDB	-17.9	-13.6	-3.9	4.8	14.6	21.6	21.7	20.7	13.5	2.3	-7.0	-16.9		
	10%	WB	-20.4	-16.8	-7.8	-0.3	7.5	14.5	16.7	15.4	8.8	-1.1	-10.8	-19.9		
		MCDB	-20.0	-16.1	-6.1	2.8	12.5	20.2	20.6	19.3	11.8	0.7	-10.1	-19.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.5	10.9	12.2	12.2	11.6	13.8	11.9	12.5	10.7	9.8	9.8	8.1		
		MCDBR	11.0	13.2	14.5	14.5	18.6	19.2	17.4	17.5	16.8	12.5	10.9	10.5		
		MCWBR	10.5	12.0	12.2	9.9	10.7	9.4	8.5	9.6	10.6	9.0	9.5	9.9		
		MCDBR	10.9	12.9	14.0	13.3	16.0	15.5	12.7	13.6	14.0	10.9	10.7	10.4		
	5% WB	MCWBR	10.4	11.9	11.9	9.1	9.7	8.8	7.2	8.2	9.5	8.2	9.4	9.8		
(40) Clear-Sky Solar Irradiance	taub		0.227	0.257	0.293	0.391	0.444	0.420	0.432	0.378	0.326	0.291	0.241	0.219		
	taud		2.122	2.125	2.079	1.884	2.001	2.223	2.211	2.359	2.449	2.294	2.112	2.048		
	Ebn at Noon		697	812	864	812	793	824	803	831	836	764	655	592		
	Edh at Noon		57	87	118	169	161	131	130	105	83	73	57	47		
(44) All-Sky Solar Radiation	RadAvg		0.96	2.07	3.90	5.55	5.13	5.50	4.89	4.08	2.92	1.86	1.14	0.67		
	RadStd		0.04	0.07	0.14	0.27	0.42	0.44	0.48	0.45	0.27	0.15	0.07	0.02		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page